

Assignment 2: Convolution – Cats vs Dogs Classification

1. Purpose

This assignment explores the use of convolutional neural networks (CNNs) in image classification and determines the effect of a training sample size on the performance of the model. It further contrasts the training of a model with no prior knowledge and using a transfer learning with a trained network.

2. Data and Study Design

The data sample is comprised of 2000 labelled images (1000 cats and 1000 dogs). In the case of Question 1, 1000 images (500 of each class) were used in training. To keep the experiments consistent, validation and test sets were set at 500 each (250 per class). The later experiments improved the size of the training sample to examine performance improvement and determine the most optimal dataset size.

3. Methodology

3.1 CNN Trained from Scratch

An original CNN architecture was designed that includes several convolutional layers that are followed by pooling layers and dense layers. A method was used to mitigate the issue of overfitting through the use of data augmentation (random flips, rotations, zoom), dropout regularization, and batch normalization. Training efficiency and unnecessary epochs were optimized by early stopping the training process and schedule the learning rate.

3.2 Transfer Learning (Pretrained Model)

Transfer learning was done using a pretrained MobileNetV2 model. At the beginning, the convolutional base was frozen to serve as a feature extractor. Afterwards, fine-tuning was conducted by unfreezing the layers of the higher-levels to enable the model to fit in the cats vs dogs data set. This method applies characteristics acquired on massive data sets like ImageNet, like edges, textures, and object patterns.

4. Results

Approach	Question	Training	Validation	Test	Accuracy
Scratch	Q1	1000	500	500	0.544
Scratch	Q2	1400	500	500	0.724
Scratch	Q3	1800	500	500	0.738
Pretrained	Q1	1000	500	500	0.978
Pretrained	Best	1400	500	500	0.986

5. Analysis and Discussion

The scratch CNN was also improving greatly with the increase in the training sample size. The start value of accuracy of 0.544 means that there is overfitting in which the model has memorized the training data, but it has not generalized. When the dataset was expanded to 1400 and 1800 images, the performance rose to 0.724 and 0.738 respectively which proves that larger datasets give more representative features and minimize variance.

Nevertheless, the pretrained model achieved much higher results compared to the scratch model in all experiments. It was effective in transfer learning as it was able to attain 0.978 accuracy using only 1000 training images. It is based on the fact that already trained models acquire generalized low-level and middle-level features like edges, contours, and textures using large datasets. These characteristics can be applied to new jobs.

Fine-tuning further increased the performance to 0.986, which means that the model can be made to focus on differentiating between cats and dogs by adjusting higher-level layers. Pretrained model also converged faster and validation accuracy was more stable indicating better optimization and less overfitting.

The size of training has a nonlinear relationship with performance. Although larger data enhances the outcomes, the rate of improvement becomes smaller with larger data. This can be observed because the difference between 1400 to 1800 images is low as compared to the past gains. This implies decreasing marginal returns to a certain size of dataset.

6. Conclusion

This paper shows that the size of data set and the model used vary greatly in terms of influencing the performance of CNN. Pretrained models always perform better because

they can use learned features when adding more training data only increases the models trained by scratch.

Transfer learning particularly comes in handy when limited data is available and this saves training time as well as enhancing accuracy. Pretrained models should be used in real-life applications because of their efficiency, scalability, and strength.

7. Graphical Results

The quantitative results are supported by the following figures that indicate the behavior of training, validation and the impact of the size of the training sample on the test accuracy.

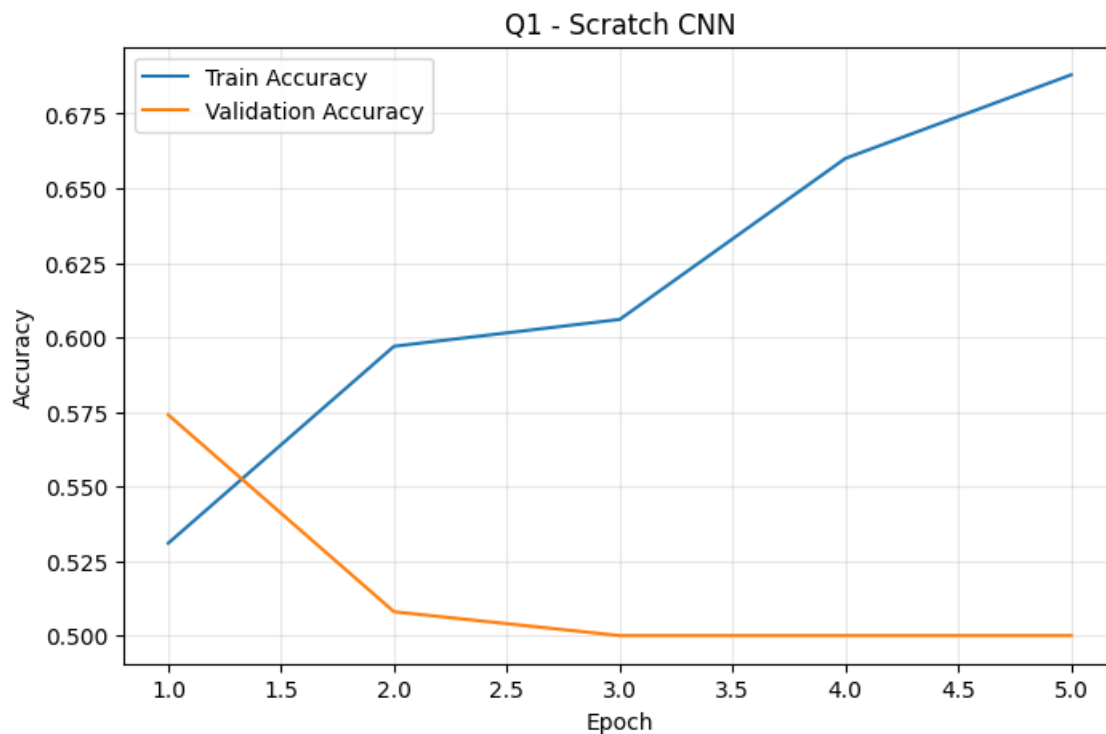


Figure 1. Scratch CNN performance for Q1 with 1,000 total training images.

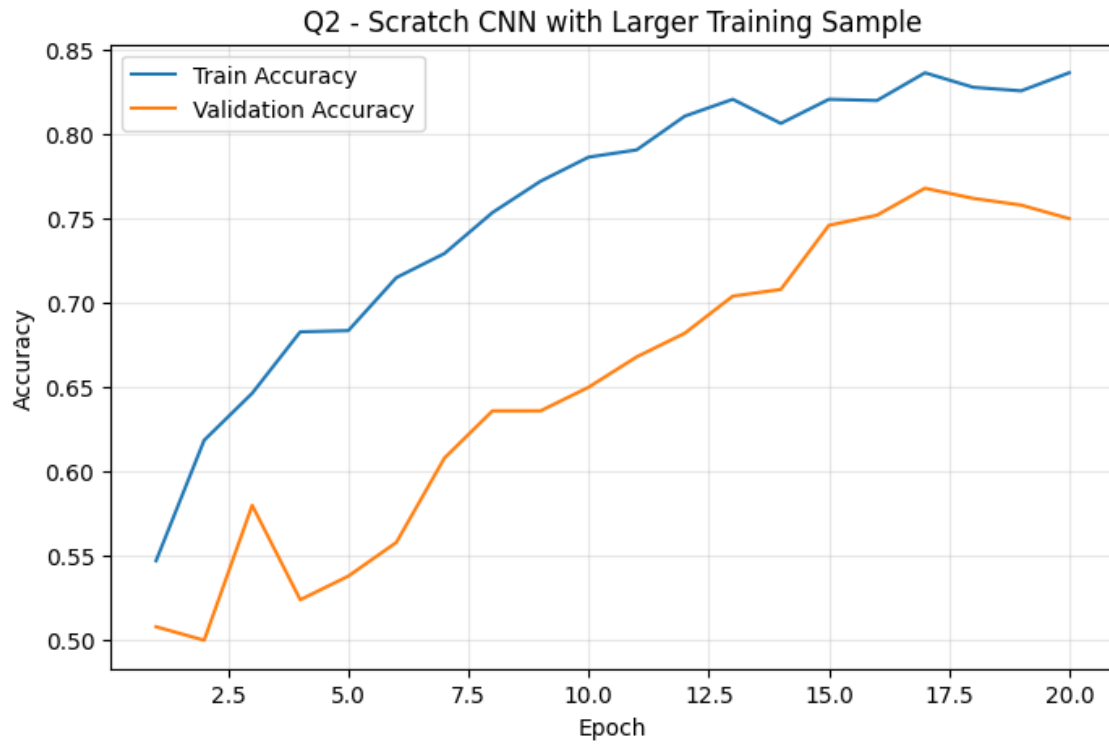


Figure 2. Scratch CNN performance for Q2 after increasing the training sample size.

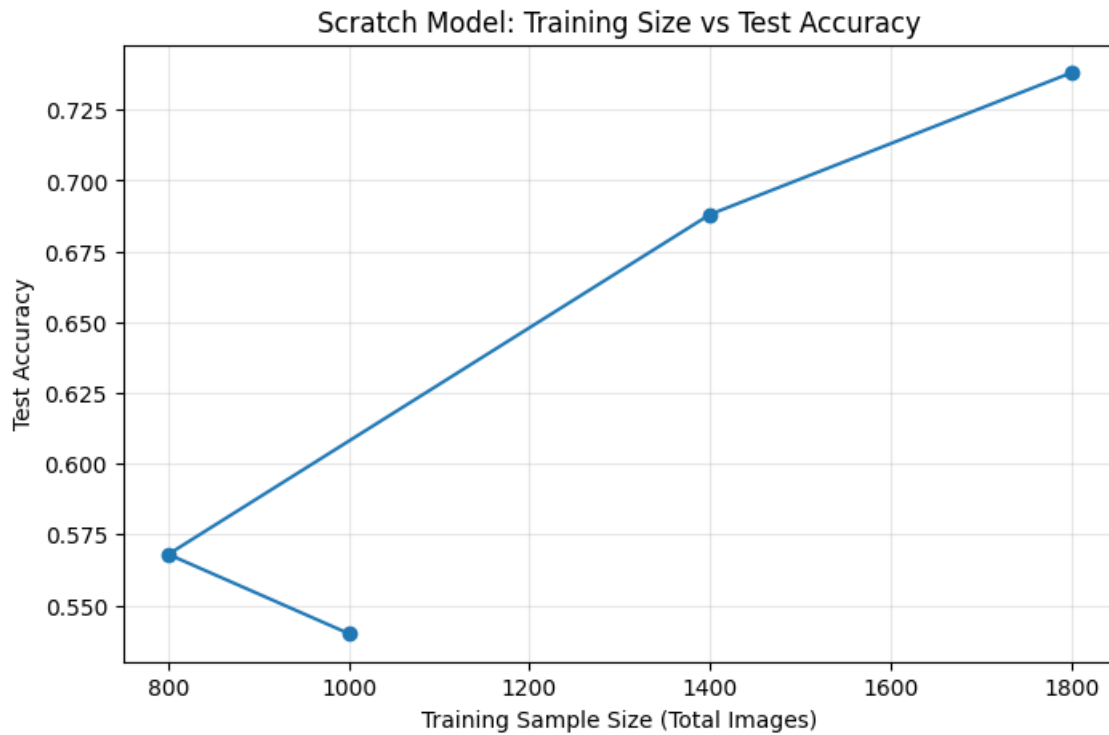


Figure 3. Relationship between scratch-model training size and test accuracy.

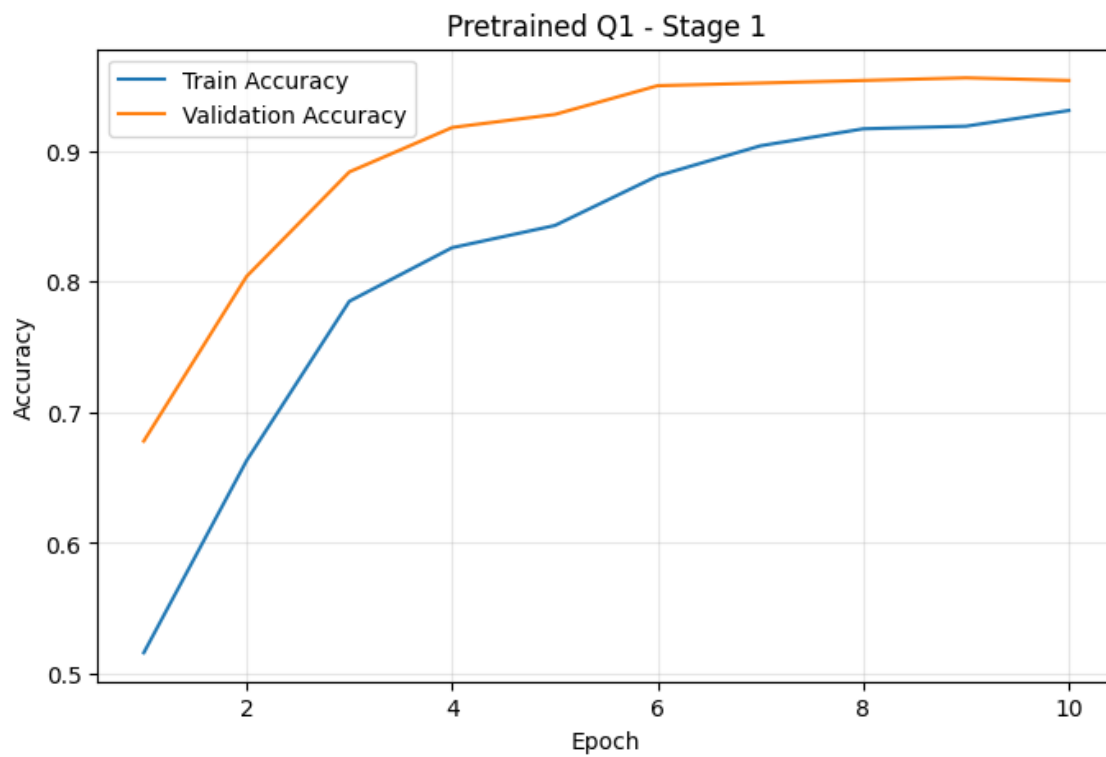


Figure 4. Pretrained MobileNetV2 Stage 1 training and validation accuracy.

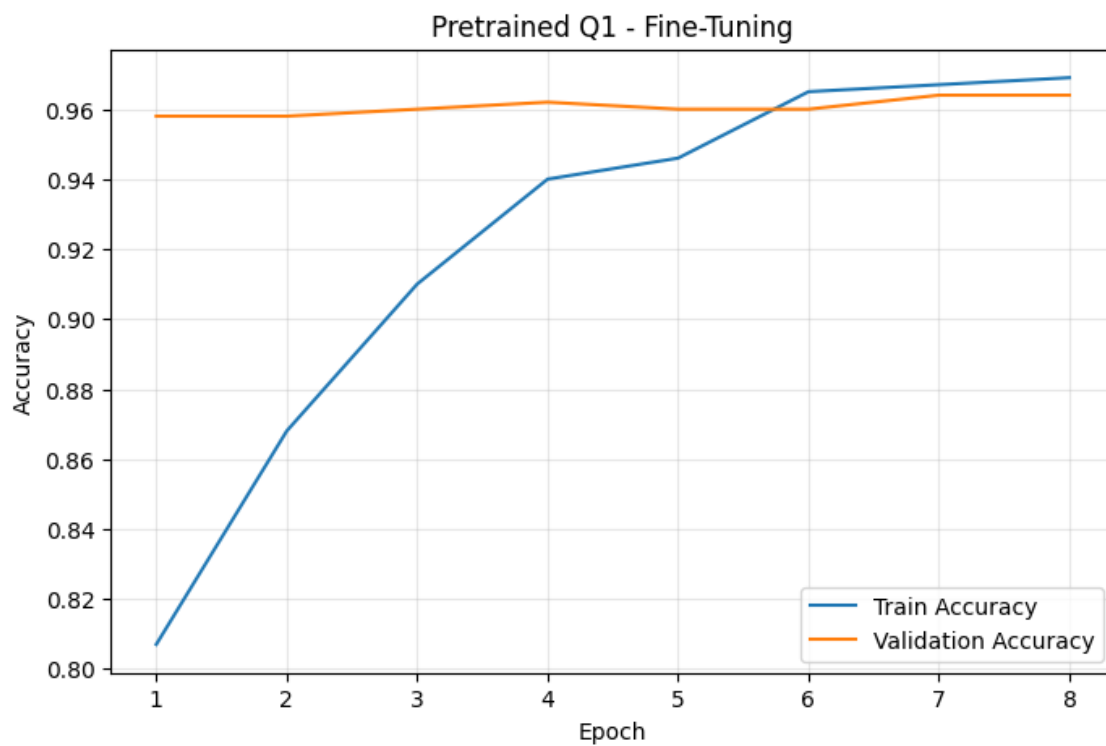


Figure 5. Pretrained MobileNetV2 fine-tuning performance for Q1.

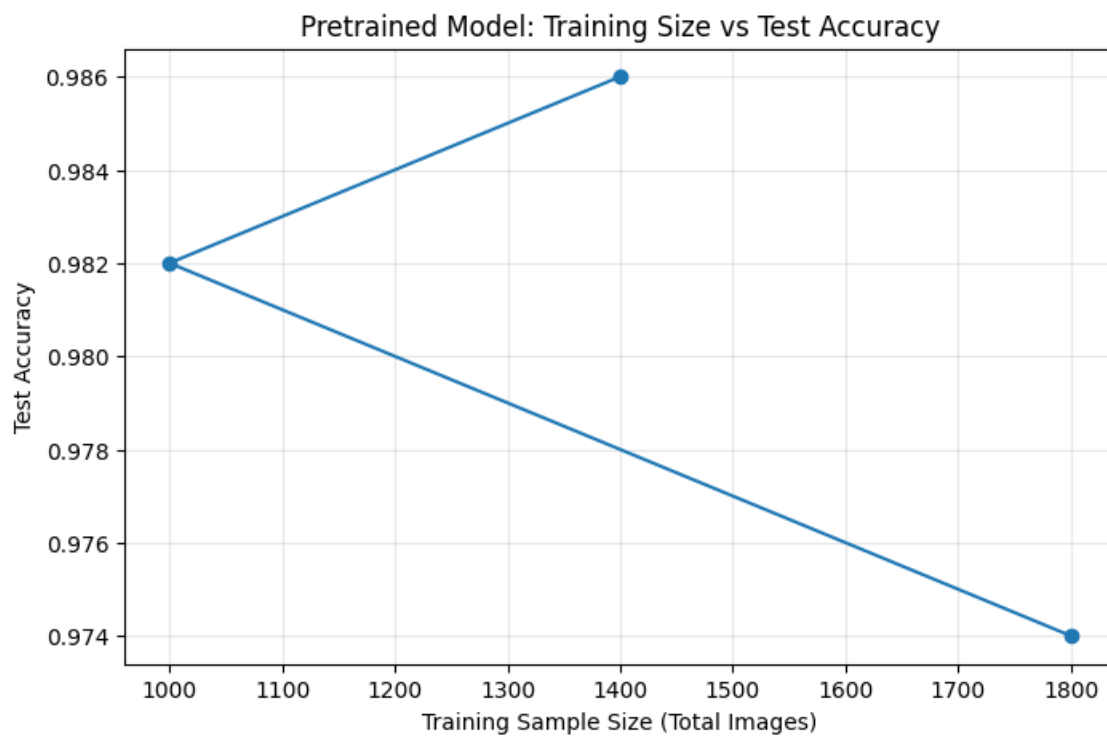


Figure 6. Relationship between pretrained-model training size and test accuracy.